

**5** Small variations in DNA sequences and activation of genes can explain why individuals respond differently to diseases and the drugs used to treat them.

(a) Describe how active genes could be identified.

(2)

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(b) Epigenetic modification can have an effect on the activity of genes in a cell.

(i) Which is the role of histones in the cell?

(1)

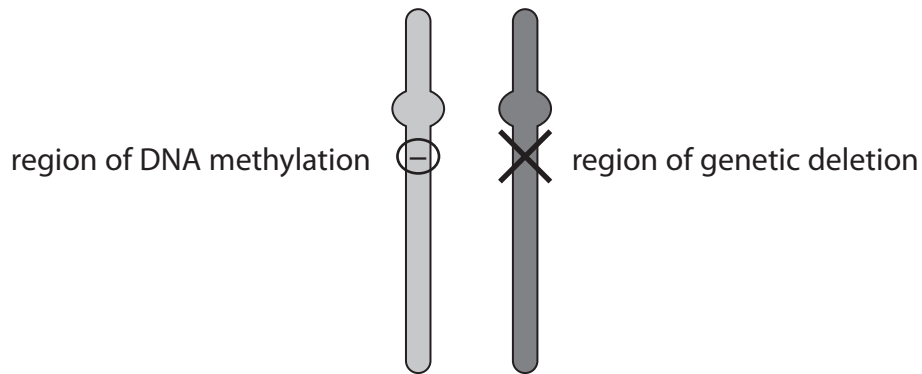
- ☐ **A** act as a transcription factor
- ☐ **B** form pores in the nuclear membrane
- ☐ **C** give structural support for DNA molecules
- ☐ **D** produce RNA polymerase

(ii) Which is the process that regulates gene expression?

(1)

- ☐ **A** crossing over
- ☐ **B** epigenetic changes
- ☐ **C** post-transcriptional changes
- ☐ **D** post-translational changes

- The diagram shows these chromosomes for an individual with Prader–Willi syndrome.



(3)

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (ii) The symptoms of Prader–Willi syndrome can be treated using human growth hormone (HGH).

This hormone is a protein produced by the pituitary gland.

Describe how *Escherichia coli* bacteria can be used to produce human growth hormone.

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(Total for Question 5 = 11 marks)